

SMART CLASS

AI INFRASTRUCTURE FOR EDUCATION

Beyond the main poster: architecture, data flow and tools

REAL-TIME EDUCATIONAL INTELLIGENCE

Edge AI | Privacy by Design | Teacher Automation | Predictive Analytics

COMPLEMENTARY ROLL-UP MODULES

This stand reveals what is not repeated on the main poster: internal processes, technical architecture, AI modes, integrations and administration tools.

EDGE / LOCAL AI

video is processed near the camera with minimal data transfer

DATA FLOW

frame to features to event to insight

TEACHER TOOLS

auto reports, prompts, event log and export

PARENT TOOLS

safe progress summaries without excess raw data

ADMIN CENTER

classes, roles, devices and access control

SECURITY LAYER

encryption, anonymization and retention policies

HOW IT WORKS



- CAMERAS**
signal from cameras and microphones
- LOCAL AI**
feature extraction on edge nodes
- EVENT DETECTION**
lesson events are detected
- ANALYTICS CORE**
attention and engagement aggregation
- TEACHER DASHBOARD**
prompts and reports for teachers
- PARENT REPORTS**
safe progress summaries

AI / ANALYTICS LAYER



Attention Heatmap

focus map by classroom zones

Emotion Trends

class mood and stress dynamics

Activity Tracking

student and group participation

Predictive Risk

early signals of disengagement

Lesson Quality Index

combined score for lesson quality

Anomaly Alerts

events that require attention

TECHNOLOGY STACK

- | | | | | |
|----------------------|--------------------|------------|-------------------|------------------|
| Computer Vision | OpenCV | NLP | Local AI | Edge Processing |
| Real-Time Monitoring | Learning Analytics | Cloud Sync | Encrypted Storage | API Integrations |



WHO IT SERVES

Teachers

less routine and more focus on teaching

Parents

clear progress dynamics, safely summarized

Students

personalized learning trajectory

Schools

quality management across classes

AI ADVANTAGE

Conventional control

manual observation, subjectivity and delayed conclusions

Smart Class AI

automation, objective data, early signals and secure analytics



CONTACTS / DEMO ACCESS

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POWERED BY

LUMINEXUS